of discriminant 5 to the canonical form $X^2 + XY - Y^2$, where s_5 is the odd root of $x^2 = 5 \bmod p$.

This reduction algorithm, outlined by Gauss and cleverly reported in Buell [3], was described in a masterly fashion in Mathews' book [13], and thus it will be referred to as Mathews' reduction. It can be applied to any quadratic form of a non-perfect square discriminant. The square root s5 may be computed with deterministic polynomial complexity by distinguishing whether p is congruent 3 or 1 modulo 4. In the first case, s_5 is one of the two square roots $\pm 5^{\frac{p+1}{4}} \bmod p$, and the computation clearly has deterministic polynomial complexity. In the second case Schoof's algorithm [18, 21] is applied to compute both $\sqrt{-1}$ and $\sqrt{-5}$ modulo p, thus s_5 is one of the two values $\pm\sqrt{-1}\sqrt{-5} \bmod p$. Schoof's algorithm evaluates a square root modulo p in a number of arithmetical operations of order $O(\log^t p)$ with $t \leq 10$, that is with deterministic polynomial complexity [18]. A description of Schoof's algorithm can be found in the original paper [18] or in Washington's book [21], and is based on properties of elliptic curves that are briefly summarized.

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1. Representation of primes in Z(ω)

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2. Computational remarks

The representations of primes $\pi \in Q(\omega)$ as the sums of two squares, established by Theorems 2 and 3, are computed with deterministic poly-nomial complexity using procedures that will be described separately for the different cases.

4.1 *Primes* $p \equiv 13, 17 \bmod 20$

The representation p = a2 + b2 does not need any further comment, since p = 1 mod 4 is the sum of two integer squares by Fermat's theorem, and its computation was shown in Section 2.

4.2 Primes p

$$+ y_0 F_{2k+1})$$

.....

5. Conclusions

The representation of primes as sums of two squares in Z(ω) allows us to formulate a theorem concerning the representation of any integer in Z(ω):

Theorem 4. *In the field Q(ω), any totally positive integer*

.....

The computation of the two-square representation of any element in Z(ω) is equivalent to factoring, as in Z, therefore at the present time only the two-square representations of primes can be computed with deterministic polynomial complexity.

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